Abdelrahman Elsaid Elsawy

Robotics and Vehicle Simulation Engineer
ROS/ROS2 | BeamNG.tech | Simulation Integration | API Bridges | Automated QA
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Summary

Robotics and software engineer with more than 6 years of experience in ROS/ROS2, robotics system integration, autonomous systems, and high-fidelity vehicle simulation. Experienced in BeamNG.tech integration, MATLAB/Simulink workflows, API bridge development, real-time simulation and data acquisition, automated QA testing, debugging, technical documentation, and customer-facing technical support. Skilled in C/C++, Python, Lua, Bash, Git, Docker, Ubuntu, Gazebo, RViz, MATLAB, and Simulink.

Core Skills

Robotics: ROS, ROS2, SLAM, MoveIt, Gazebo, RViz, autonomous navigation, sensor fusion
Simulation: BeamNG.tech, vehicle simulation, high-fidelity simulation, simulation workflows
Programming: C, C++, Python, Lua, SQL, Bash
Integration: API bridges, system integration, real-time simulation, data acquisition, MATLAB, Simulink
Testing and QA: automated QA testing, regression testing, bug reproduction, debugging, root-cause analysis
Tools: Git, Docker, Ubuntu, Zendesk, technical documentation

Professional Experience

Research Software Engineer
Nov 2021 – Present
BeamNG GmbH
Remote

- Integrated BeamNG.tech with ROS/ROS2, MATLAB, and Simulink for high-fidelity vehicle simulation and external robotics workflows.
- Developed and maintained API bridges for real-time simulation control, data acquisition, vehicle interaction, and external tool integration.
- Performed automated QA testing, regression analysis, and issue investigation using C++, Python, and Lua.
- Investigated customer-reported and internally detected issues by reproducing bugs, analyzing root causes, and providing fixes, workarounds, or technical guidance.
- Supported global B2B users through Zendesk and technical forums on simulation, APIs, integrations, and workflows.
- Improved API documentation, tutorials, integration examples, MATLAB/Simulink guidance, and developer-facing technical materials.
- Converted recurring QA and support issues into reusable test cases, documentation improvements, and actionable product feedback.
- Contributed to BeamNG.tech engineering repositories, including BeamNGpy, API documentation, simulation workflows, QA fixes, tutorials, and integration examples.
- Implemented and validated scenario and mission-loading workflows, including scenario discovery, scenario loading, vehicle retrieval, sensor attachment, vehicle connection, scripted control, and sensor polling.

Mechatronics Systems Engineer
Feb 2020 – Jul 2021
Ideasparq Robotics Sdn Bhd
On-site

- Developed ROS-based control systems and autonomous navigation solutions for service robots.
- Implemented sensor fusion algorithms for robotics applications.
- Tested robotic platforms using Gazebo, RViz, Docker, and Ubuntu.
- Worked on system integration across software, sensors, control logic, simulation, and deployment environments.

Autonomous Robotics Engineer
Jun 2019 – Jan 2020
ENDSTRUCT
On-site

- Developed autonomous navigation systems for GOVTECH mobile robots.
- Developed health monitoring systems for mobile robotic platforms.

Research Assistant
Sep 2019 – Dec 2019
Autonomous Vehicle Laboratory
UMPSA Universiti Malaysia Pahang Al-Sultan Abdullah

- Processed 3D LiDAR data for real-time object detection.
- Developed ROS-based middleware for sensor abstraction.

Selected Technical Contributions

- BeamNG.tech integration with ROS/ROS2, MATLAB, and Simulink.
- BeamNGpy and API bridge development for real-time simulation control and data acquisition.
- Scenario and mission-loading validation for automated simulation workflows.
- Automated QA, regression testing, bug reproduction, debugging, and root-cause analysis.
- Technical support for global B2B users through Zendesk and technical forums.
- Documentation and integration examples for APIs, simulation workflows, MATLAB/Simulink, and HIL/DIL material.
- Robotics system integration using ROS, Gazebo, RViz, Docker, and Ubuntu.

Education

Bachelor of Engineering in Mechatronics, Robotics, and Automation Engineering 2016 – 2020
UMPSA Universiti Malaysia Pahang Al-Sultan Abdullah

Publication

The Flexural Strength Prediction of Porous Cu-Sn-Ti Composites via Artificial Neural Networks
The 6th International Conference on Robotics Intelligence and Applications, 2018
DOI: 10.1007/978-981-13-8323-6_34

Courses and Workshops

- Learn ROS2 as a ROS1 Developer and Migrate Your ROS Projects – Udemy, Nov 2022
- Level 2: Simulation and Hardware Implementation for SLAM – Udemy, Nov 2022
- Getting Started with Azure DevOps Boards – Coursera, Oct 2022
- Command Line in Linux – Coursera, Sep 2022
- Arduino Workshop – Core Electronics, Sep 2022
- Open-Source Software Development Methods – The Linux Foundation, Oct 2021
- Python Scripting for DevOps Specialization – LearnQuest, Sep 2021
- Python Scripting: Files, Inheritance, and Databases – LearnQuest, Sep 2021
- Web Application Technologies and Django – University of Michigan, Sep 2021
- AWS Cloud Technical Essentials – AWS, Aug 2021
- DevOps and Build Automation with Python – LearnQuest, Aug 2021
- Python Scripting: Dates, Classes, and Collections – LearnQuest, Aug 2021
- Configuration Management and the Cloud – Google, Aug 2021
- Introduction to Git and GitHub – Google, Jul 2021
- Technical Support Fundamentals – Google, Jul 2021
- Troubleshooting and Debugging Techniques – Google, Jul 2021
- Using Python to Interact with the Operating System – Google, Jun 2021
- Crash Course on Python – Google, May 2021
- Malaysia Autonomous Vehicle Challenge (MAVeC'19) – MARii, Nov 2018
- The International Conference on Robotics Intelligence and Applications (RiTA 2018) – KIAST SG, 2018
- Boundless Innovation Week – HPMM, Aug 2018
- CAM Workshop – The Institution of Engineers, Malaysia, Nov 2017
- IEM MATLAB Workshop – The Institution of Engineers, Malaysia, Oct 2017
- Arduino for Beginners Workshop – SPARKTECH Team at IIUM, Nov 2016
- Online Database Class: SCOPUS – University Malaysia Pahang, Nov 2016

Languages

Arabic: Native or bilingual proficiency
English: Professional working proficiency